



School of Chemistry

Third Year

Module CHEM 30110

Instrumental Analysis

Experimental Practical Course

Laboratory Manual

List of Experiments

Experiment 1: Determination of the Quinine concentration in Tonic Water Samples Using Fluorescence Spectroscopy

Experiment 2: Use of Atomic Absorption Spectroscopy to Determine the Trace Metal ion concentration of Calcium and Magnesium

Experiment 3: Analysis of Alcoholic Drinks by Gas Chromatography

Experiment 4: Analysis of Soft Drinks by Liquid Chromatography

SAFETY CONSIDERATIONS

Before proceeding to bench work, you must sign the declaration provided on page 11, acknowledging that you have read and understood the safety material provided.

- Laboratory hygiene and good “housekeeping” are important aspects of Health & Safety.
- Be clean and tidy on the bench, work carefully and attend immediately to spills of any nature.
- Always leave the bench in a state you should like to find it: neat, tidy, clean and dry.

Failure to clean up after the experiment performed will result in lost of marks.

Safety Information

Mandatory Reading

Under the Safety Health and Welfare at Work Act 1989, all those involved in a given enterprise, be it employment, education, etc., have a statutory duty to take care of themselves and others who may be affected by their acts or omissions. In compliance with this, the following guidelines are provided for students.

ALL STUDENTS HAVE THE FOLLOWING RESPONSIBILITIES WHILE PRESENT IN THE LABORATORY:

1. Students must co-operate with staff in the wearing of the correct safety equipment, using the appropriate safety devices and following proper safe systems of work. Undergraduates are required to provide their own laboratory coats and other Personal Protective Equipment as deemed necessary by the Head of Department.
2. All working areas must be kept clean and tidy, and high standards of housekeeping and hygiene maintained.
3. Students must co-operate in the investigation of accidents and the reporting of them and also the reporting to their supervisors of any local hazards of which they become aware.
4. All students will be encouraged to promote ideas on the improvement of health and safety standards and also provide suitable suggestions for reduction in risks.
5. Students must not interfere with or misuse any specified items of safety equipment/device.
6. Students are advised that strict requirements under the Safety Health and Welfare at Work Act 1989 can be used by the enforcing authorities against persons found guilty of reckless behaviour.

Undergraduate Students

In addition to their general obligations outlined above, all *undergraduate* students have the following responsibilities:

- They must attend any relevant safety instruction talks before carrying out experiments.
- They must carefully follow instructions contained in the laboratory manuals, and given by demonstrators and supervisors.

PROCEDURES IN THE EVENT OF FIRE, ACCIDENT OR INJURY

FIRE

All users of the building must familiarise themselves with the nearest exit route from their working area. Be sure you know the location of the nearest fire extinguisher, body shower and fire blanket.

IN THE EVENT OF FIRE

Remember the first priority is for personal safety: if in doubt, get out!

1. If possible, without any personal risk, close windows and doors, switch off and close fume cupboards, and switch off electrical sources and experiments in the near vicinity, but **leave lights on.**
2. Tackle small bench fires with the fire-fighting equipment available, but **do not take personal risks.**
3. Where possible, remove gas cylinders from the vicinity.
4. If in any doubt that the fire may get out of control, or the bench fabric becomes ignited, **sound the fire alarm.**

EVACUATION PROCEDURE

When the fire alarm sounds:

1. Leave the building immediately by the nearest available exit. On leaving, go directly to the designated Assembly Point (square outside new Students' Centre).
2. Never use lifts in the event of a fire alarm.
3. Technical staff controls the entrance to the building during an emergency; re-entry will not be allowed until the Senior Fire Officer present give permission.
4. After the fire alarm has sounded, the College Technical staff will take all necessary action until the arrival of the fire brigade's Senior Officer, present on the appliance, assumes responsibility for dealing with the fire.
5. The person or persons responsible for starting and/or discovering the fire must be available to consult with the fire brigade.

UNDERGRADUATE LABORATORY SAFETY

1. SAFETY INSTRUCTIONS

Safety signs & instructions from demonstrators and academic supervisors must be strictly obeyed.

2. EATING, SMOKING AND DRINKING

These activities are strictly forbidden in all laboratory areas or in adjacent rooms.

3. SUPERVISION OF LABORATORY WORK

No one is to perform experimental work unless a demonstrator or a member of the academic staff is present in the laboratory.

4. EYE PROTECTION

All persons must wear approved safety spectacles, goggles or visors in chemical laboratories and other designated areas. You will NOT be allowed to work without safety glasses!!!

5. PROTECTIVE CLOTHING

All persons carrying out experimental work must wear the appropriate protective clothing.

6. SAFETY AUDITS

Experiments must not be performed until a safety audit has been carried out.
Safety audits for most undergraduate experiments are available in the laboratory.

7. TRANSPORT OF CHEMICALS AND GLASSWARE

No chemicals or glassware may be carried *via* the main stairs or passenger lifts.

8. WASTE MATERIALS

Waste chemicals, incl solvents, must be placed in the appropriate waste containers for safe disposal.

9. LABELLING CONTAINERS

All containers of non-commercial chemicals must be clearly labelled showing the name of the chemical or formula, date, user's name and safety precautions, if hazardous.

10. COMPRESSED GAS CYLINDERS

Compressed gas cylinders must be secured, either with strap or chain, at all times. Cylinders should only be moved in a cylinder trolley. Cylinder control-valves must be removed prior to moving.

11. UNATTENDED APPARATUS/EXPERIMENTS

If equipment is in continuous operation, or experiments are left unattended overnight, two copies of the appropriate form must be completed and signed by the academic supervisor. One copy is to be left beside the apparatus and the other placed in the box provided on the laboratory door.

12. REPORTING ACCIDENTS, FIRE AND DANGEROUS OCCURRENCE

All accidents resulting in injury, property damage or fire, or any dangerous occurrence that might have caused injury, must be reported immediately to a demonstrator, laboratory technician or academic staff. In the interests of full disclosure, no penalties normally accrue to such reporting.

RISK ASSESSMENT: THE USE OF GLASSWARE

Cuts from broken glass or the misuse of glass remain amongst the commonest form of injury in any Chemistry School/Department. *Great care is always required.*

Hazards

- Cuts from damaged or broken glass.
- Cuts from flying glass due to explosion, implosion, evacuation or mechanical shock or stress.
- Cuts from forcing tubing, teats or bungs onto glass tubing, pipettes or condensers that then break.
- Cuts from broken glass and sharp items e.g. Pasteur pipettes disposed in ordinary waste bins.
- Burns from heated glass.
- Poisoning following cuts by contaminated glassware.

Precautions

- **Before use, check all glassware is free from cracks, flaws/scratches that may cause it to fail.**
- Have damaged glassware repaired or dispose of it in the "Broken Glass / Sharps" bin. Do not use the ordinary waste bins. Use a brush and dustpan to clear up broken glass. Be especially careful with broken glass in sinks where the sharp edges may be invisible. Use tongs to pick out pieces.
- When fitting to tubing, lubricate the glass with water or glycerol and soften plastic tubing by brief immersion in hot water. Do not use excessive force. **Do not exert force in a direction, which will make the glass snap.** Plan where the sharp edge of the glass might go on breaking and arrange your grip accordingly. Wrap the glass in a towel or thick layers of paper tissue. In removing tubing, use carefully a sharp knife to cut through tubing that does not come off easily.
- Place **hot glass** where no one can accidentally come into contact with it before it has cooled.
- **Joints and stoppers.** Lubricate ground glass connections before assembling and disassemble them immediately after use. Do not stopper hot flasks or containers. If a stopper seizes, do not reheat the container to remove it.
- **Vacuum or pressure use.** Inspected for flaws before use.
 - ◇ **Pressure:** Use only special glassware rated well above the pressure to be used.
 - ◇ **Vacuum:** For glassware under vacuum, volumes of 1 litre or larger should be enclosed in tape or plastic mesh to restrain fragments in the event of implosion. This applies to equipment such as vacuum storage bulbs, rotary evaporators, vacuum dessicators, etc.
- **Washing.** Detergents are the normal means of cleaning. More drastic methods (e.g. base baths or chromic acid baths should be used only when detergents or solvents are inadequate and should be covered by a separate Risk Assessment. Beware of fire risk if using solvents to clean or dry.

Emergency Procedures

Treat cuts and burns immediately. Apart from very minor injuries, call for First Aid treatment.

Specific Cautions

- **Attaching pipettes to fillers.** The coloured band on the neck of the pipette is a quality mark, and does NOT indicate the distance for insertion into the filler. Pipettes need to be inserted only approx. 1 cm into the holder. Hold the pipette close to the top of the neck and insert gently into the holder.
- **Stoppering volumetric flasks.** Do not force or hammer the stopper into volumetric flasks.

RISK ASSESSMENT: USE OF STANDARD ELECTRICAL EQUIPMENT

Laboratory supply of mains electricity is via individual socket outlets, which may be 3-phase 415 V, 50 Hz, or single-phase 240 V, 50 Hz. Office supplies are normally 240 V, 50 Hz. Most electrical equipment operates on 240 V, 50 Hz.

A very wide range of electrically powered equipment is found in the laboratory including fluid and vacuum pumps, lasers, power supplies, electrophoresis and electrochemical apparatus, X-ray equipment, stirrers, hot plates, ovens both conventional and microwave and computers, printers and video equipment.

In the office, there are, amongst other things, computer equipment, FAX machines and photocopiers. In the lecture theatres there are overhead and slide projectors. In fact, everyone in the School of Chemistry is exposed at some time to electrical equipment.

Hazards

- Electric shock is the effect produced on the body and particularly on the nervous system by an electrical current passing through it. The effect depends on the current strength which itself depends on the voltage and body resistance i.e. path length and surface resistance of skin (which is much reduced when wet). Death can be the result of the normal voltage of 240 V causing currents of greater than 30 mA to flow through the body for more than 40 ms. Minor shocks may also cause injury following involuntary muscle contraction.
- Burns caused by the passage of heavy currents through the body or by direct contact with an electrically heated surface.
- Explosion and fire caused by electrical sparks, short circuits or overload heating, old wiring in the presence of flammable material.
- Injury from microwave and radiofrequency sources and from induction heating equipment.

Precautions

The following precautions are not meant to be exhaustive or to cover aspects of repair or construction of electrical equipment but to cover everyday use in laboratory and office.

Plugs and fuses

- Do not use plugs that are cracked or broken. Ensure that plugs are wired properly, that conductors are securely fixed and that the cable is firmly held by the strain relief grip.
- Ensure that the rating of the fuse is appropriate to the appliance. Most electronic equipment (computers, measuring instruments etc.) requires only a 3 A fuse which will load to 720 W.

Reserve 13 A fuses (loading to 3000 W) for heavier equipment. If in doubt check with the Electronic Workshop personnel.

Cabling

- Ensure that cable is in good condition and free from breaks in the insulation. Cable must be sufficiently robust to withstand the wear and tear of laboratory or office use and fully waterproof where water may come within the vicinity of the apparatus.
- Cables must not be run across the floor in such a way as to cause a tripping hazard or to be susceptible to damage from passing traffic. If it is necessary to run cables across walkways, cover them with cable protectors.

Extensions

It is permissible if necessary to feed one four way extension block from a single socket provided the block feeds only low power equipment (less than 500 W or 2 A). Do not daisy-chain extension leads. Kettles, microwaves and heaters, which have higher power demands, must not be used on an extension but fed from an installed socket point.

Mains Switch

Make sure you know where the mains switch location so that you can turn off power in an emergency.

Use

- No apparatus with exposed mains terminals should ever be used.
- Do not use ordinary electrical equipment in the vicinity of flammable or explosive gases. Ordinary electrical equipment could be a source of ignition.
- Likewise do not use ordinary electrical equipment where it may get wet. Water may cause a dangerous short circuit.
- Never switch on equipment that has had liquid spilt on it until the equipment has been tested. Tell anyone to whom you take the equipment for testing what has happened.

Repairs

Do not attempt to repair electrical equipment by yourself unless you are competent to do so. Ensure that the equipment is disconnected from the main power before beginning.

If in doubt, take equipment to the Electronics Workshop for repair.

Risk Remaining

Mains electricity will always remain a potentially lethal hazard if mishandled. Following the precautions outline above, the risk remaining should be minimal.

Emergency Procedure for Electric Shock

Switch off the power before touching the injured person. Contact First-Aid personnel.

RISK ASSESSMENT: USE OF CHEMICAL SUBSTANCES

Hazards:

Many chemical substances are potentially hazardous in a variety of ways.

Precautions

The precautions necessary vary from substance to substance and should always be reviewed whenever a substance that is new to the user is encountered.

Material Safety Data Sheets (MSDSs)

It is important that information about the potential hazards of chemicals is widely available. To this end, MSDSs are now readily accessible from computer networks, such as UCD's library database and the internet, and from manufacturers (Aldrich or Fisher). A book of MSDS for all compounds described in this manual are located in a book located in the laboratory. Thus, if you buy a chemical to use in a laboratory, ask the manufacturer for a complete MSDS sheet. The sheets contain numerous abbreviations and codes, but glossaries are available. For example, the LD50 is defined as: ***LD50: (Lethal Dose 50) is the quantity of a chemical that kills 50% of a sample population.*** In full reporting, the dose, treatment and observation period should be given. Further, LD50, LC50, ED50 and similar figures are strictly only comparable when the age, sex and nutritional state of the animals is specified. Nevertheless, such values are widely reported and used as an effective measure of the potential toxicity of chemicals.

CHEM30110 - Safety Declaration Form 2025

I agree to work in accordance with School policies, follow the procedures as described in this manual and follow the instruction of my demonstrator.

Personal Protective Equipment (PPE)

- Lab coats and safety glasses must be worn at all times while in a lab.
- Other PPE may be required; they will be supplied by your demonstrator and must be used as directed.
- Contact lenses must not be worn in synthetic chemistry laboratories.

Good laboratory practices (GLP)

- Follow procedures as written in this manual or as directed by your demonstrator.
- Keep your work area as clean and tidy as possible while working.
- Label all solutions and samples so they can be identified.
- Do not eat or drink in any lab.
- Leave the lab as you found it.

Glassware

- Use Good Laboratory Practices when handling glassware.
 - If you are not trained in GLP, follow the directions of your demonstrator.
- Report any damaged glassware to your demonstrator.

Equipment

- Use all equipment as directed by your demonstrator.
- Respect all equipment; do not misuse or tamper with any equipment.
- Report any problems or suspected faults to your demonstrator.

Wastes

- Dispose of all wastes as directed by your demonstrator.

Risk assessment

- Read the risk assessment for each practical (before starting your work).
 - Be familiar with the hazards specific to that practical and the emergency responses.

If you are pregnant, or suspect you may be pregnant you **must** inform a member of staff

Emergencies

- Report any injury or accident/incident to your demonstrator.
- Follow the evacuation instructions of demonstrators/technical staff.

Name (Block Capitals): _____

Signature: _____

Date: _____

For additional information regarding safety in the School see: <http://www.ucd.ie/chem/study/ourfacilities/healthandsafety/>



CHEM 30110 – Experiment 1

Determination of the Quinine concentration in Tonic Water Samples Using Fluorescence Spectroscopy

Learning Objectives

Understand the sensitivity of Fluorescence for analysis by determining the limit of detection for quinine.

Understand the requirements for generating a calibration curve.

Be able to use spreadsheets to manipulate data and plot spectra and graphs, including working with and expressing uncertainties.

Aim: to use fluorescence to:

- (1) Generate a fluorescence calibration curve using standards of known concentration.
- (2) Determine the Limit of Detection (LOD) and limit of Quantification (LOQ) of quinine.
- (3) Determine the concentration of quinine in commercial tonic waters.
- (4) Determine the uncertainty in the concentration of quinine.
- (5) Investigate the impact of an interferent in analytical measurements

Quinine

Quinine is an organic compound found in the bark of the cinchona tree. Historically, quinine/cinchona bark was used to prevent and treat the disease malaria. The plant species was reported in 1784 and later in 1791 its use in treating fevers was discovered in India by William Roxburgh, (English East India Company). Quinine is a key ingredient of tonic water, which was an accessible way to administer quinine as an antimalarial. The EU sets a safety limit of 100 mg/L of quinine in tonic water and soft drinks.

Quinine absorbs in the UV at 350 nm and emits in the visible spectrum at 450 nm. It is highly emissive in acidic solution, but its emission is quenched in alkaline solution and in the presence of chloride anions. Therefore, in order to improve the sensitivity of detection the analysis of quinine can be performed in acidic solution.

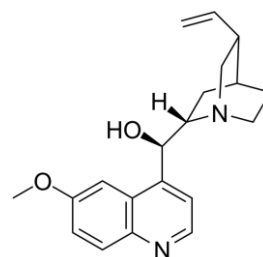


Figure 1: Molecular structure of quinine (left). Image of tonic water under visible and UV light (right).

Background Fluorescence

Fluorescence spectroscopy is a highly sensitive technique for the detection of emissive molecules. A key advantage of the technique is that unlike absorption spectroscopy it does not rely on subtracting one signal from another, but simply detects the intensity of the photons emitted from a molecule or atom as it returns from the excited state to the groundstate. One property of fluorescence is that the emitted photon has lower energy (longer wavelength) than the excited photon. This results in a separation of the wavelength of emission from the wavelength of absorption and this difference in wavelength is known as the Stokes Shift. Fluorescence is also sensitive to the environment and for many molecules pH and solvent can greatly impact the emission intensity.

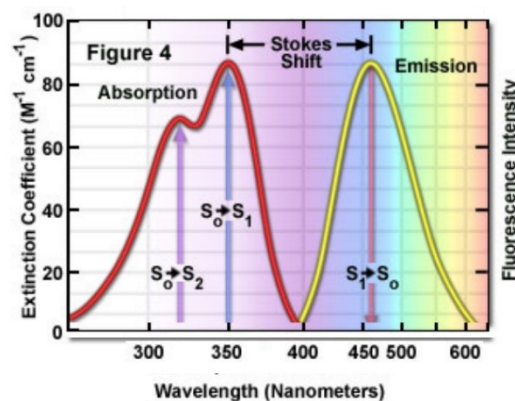


Figure 2: Absorption and emission of energy

Part 1: Constructing a calibration curve using standard solutions and determining the limit of detection (LOD) and the limit of quantification (LOQ) of quinine.

A calibration curve is an essential tool in analytical chemistry as it provides the relationship between instrument response and the amount of analyte present. A calibration curve is constructed by determining the instrument response to a set of **standard solutions** containing known amount of analyte, see Figure 3. In general, we assume that errors in the instrument response values (due to random variation) are greater than those in the values assigned to the standards. This typically involves the use of 6 standards, whose concentration should be evenly spaced over the detection range of interest.

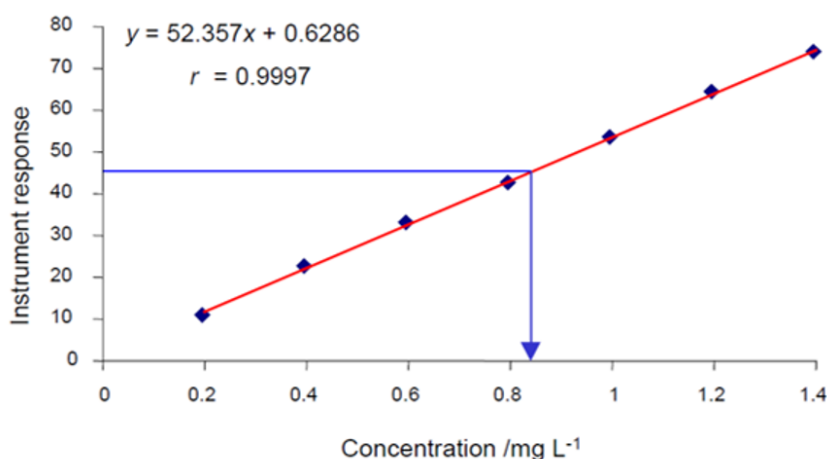
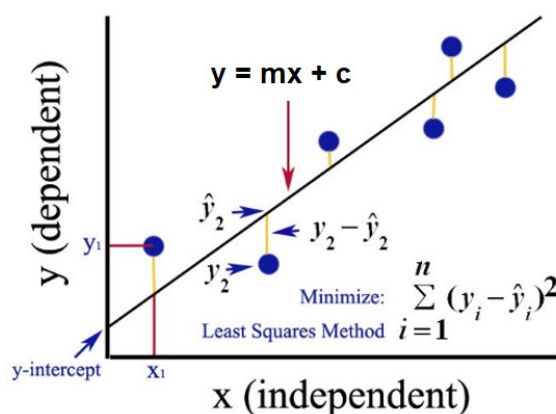
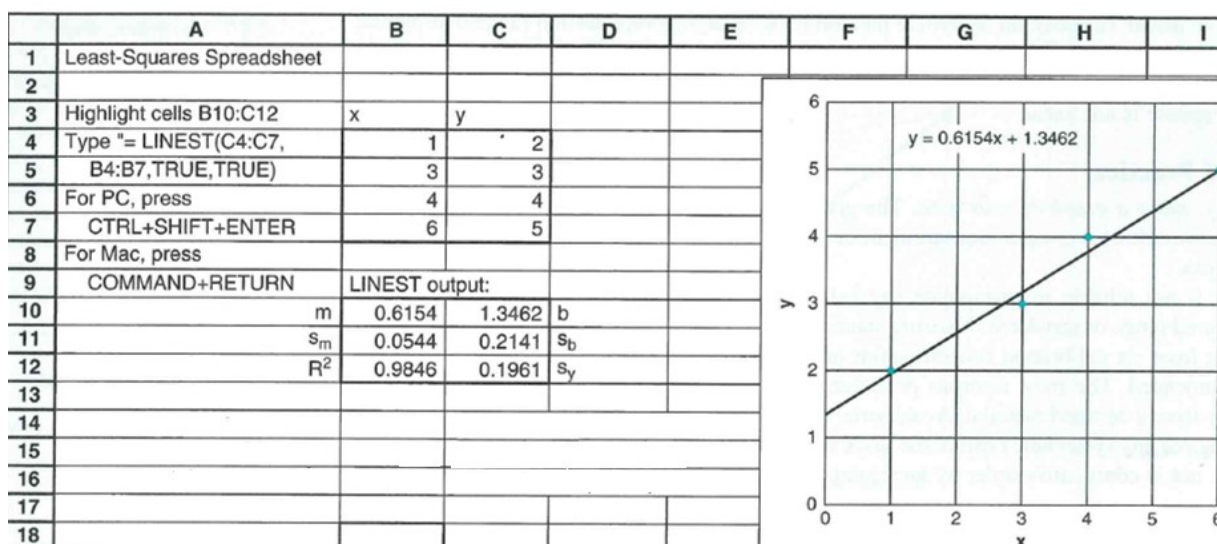


Figure 3: Example of a standard curve showing instrument response to analyte concentration.

Once the data is plotted the **line of best fit** that provides the **best correlation** with the standards is obtained. This is done using a method of mathematical **regression** (regression analysis in excel) known as the method of least squares, which obtains the best fit of the data by making the total of the square of the errors in each point as small as possible (see figure). In doing this the equation of the line that is obtained can be used to determine the



concentration of any sample based on the best correlation of the response to concentration. The **LINEST** function in excel **calculates the statistics for a line by** using the least squares method to calculate a straight line that best fits your data.



m = absolute value of the slope	b = intercept
sm= standard deviation in slope derived from LINEST	Sb = standard deviation in intercept from LINEST
R ² = correlation function – goodness of fit	sy = standard deviation of y response derived from LINEST

Figure

4: Creating LINEST output for x and y data in excel.

Limit of detection LOD and limit of quantitation LOQ

In addition to providing information about the concentration of samples the calibration curve can provide details about the detection limit, or the **limit of detection (LOD)**, which is as the lowest concentration

of analyte that is significantly different to the blank. The LOD can be estimated from the calibration curve using the following relationship

$$\text{LOD} = 3.3 s_y / m$$

Where s_y is the standard deviation of the response of a low concentration samples and is determined from either (i) the standard error of the calibration curve or (ii) the *standard deviation of the y-intercepts of the regression lines*. And m is the slope of the calibration curve. In this experiment we will use the standard deviation of the response of the curve (s_y), which will be obtained by using the LINEST function to treat the data in excel. In practice the LOD represents a signal determined from a low concentration sample is roughly $3 \times$ noise, and while it is detectable it is too low for accurate measurement. The threshold for accurate measurement is defined as the **limit of Quantitation** (LOQ, or QL, the quantitation limit).

$$\text{LOQ} = 10 s_y / m$$

Note: We are using s (for the standard deviation of the response of the curve) σ can also be used. As we are using an uncertainty value extracted from analysis, good practice is to quote this value to one significant figure and include the additional as a subscript (e.g 0.4₂ ppm).

Part 2: Determine the concentration of quinine in two different tonic water samples

In the second part of the experiment, you will use the calibration curve to determine the quinine concentration and the uncertainty in the value for dilute solutions of different commercially sourced tonic waters.

Part 3: Investigation of possible interference

In spectroscopic analysis an interferent is a substance that changes the signal while the analyte concentration remains the same. In fluorescence common interferents take the form of quenchers, which act to interact with the excited state and accept the energy or the electron and prevent emission. Thus, reducing the signal from the analyte. Chloride ions can act to quench emission and in the final part of the experiment you will monitor the impact of the addition of chloride, in the form of NaCl to the emission of quinine.

Notes on Error Analysis

Uncertainty in a value can be represented by the standard deviation (s) of the mean ($\bar{x}$).

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$
$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n-1}}$$

When recording replicate measurements this treatment does not consider the number of replicates that were taken.

Standard Error (SE) is a useful concept when multiple measurements are performed.

$$SE = s/\sqrt{n}$$

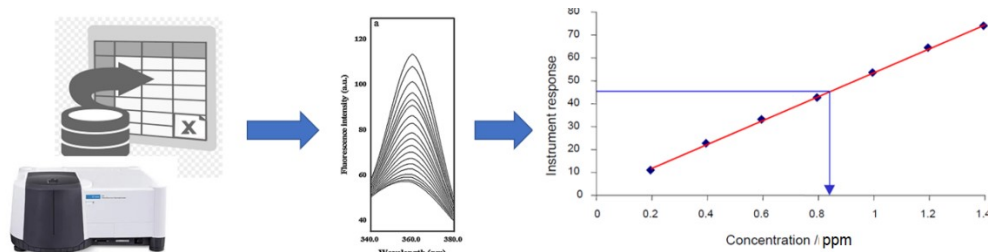
where s is the standard deviation and n is the number of samples at each concentration (here $n = 3$). This value will be less than the standard deviation.

Materials and Method

Materials and Equipment

Stock solutions + glassware required per group	Instrument settings
Range of stock solutions	
2 ppm Quinine sulphate stock x 3	Excitation = 350 nm
0.05 mol L ⁻¹ H ₂ SO ₄ solution	Emission = 380-580nm
Deionised water wash bottle	Excitation slit width = 2.5 nm
1 x 250 mL or 400 mL Beaker for waste washing	Emission slit width = 5 nm
1 x 1 mL graduated pipette	Medium scan speed = 600 nm / min
1 x 5 mL graduated pipette	
1 x green pipette pump	
14-18 x 25mL volumetric flasks	
1 x fluorescence cuvette	

Methods



Part 1: Objective to construct a calibration curve and determine the limit of detection

1. Turn on the instrument and set the settings to the correct parameters mentioned above.
2. Measure the emission of the blank solution 0.05 M H_2SO_4 . (record the emission intensity at the maximum).
3. Label the three stocks A-C.
4. Using the stock solutions provided prepare “three sets” of 6 standard solutions in the range 0.02 to 2 ppm. You should ensure that these concentrations are evenly spaced (e.g. 0.02, 0.04, 0.08, 1.2 and 1.6 ppm), and this may be done by serial dilution. This is done by pipetting a quantity of the stock solution into the 25 mL volumetric flask and making the volumetric flasks up to the mark using the 0.05 M H_2SO_4 provided.
5. You will prepare 18 solutions in total. Keep track of the parent stock by labelling these A-1 to A-6, B-1 to B-6 and C-1 to C-6. ***It is important to keep track of the samples so take care to label them.***
6. Take the emission spectrum and note the emission intensity at the maximum (ca 450 nm) for set A and then repeat this for sets B and C.
7. Make sure to save your data periodically, this can be done via the file tab on top left of the software. Always save as a .csv file so it can be easily imported for your data analysis. Be sure to keep track of the sample.

Part 2: Determination of the concentration of quinine and the uncertainty in the value

1. Prepare two tonic assay samples by diluting each tonic water 1/100 times.
2. 25 mL of the above solutions should be prepared.
3. Record the emission intensity of each sample.

Part 3: Investigate the impact of an interferent in analytical measurements

1. Prepare a 1 ppm quinine solution in 0.05 M H_2SO_4
2. Weigh out enough NaCl to add to prepare 25 mL of a 1 ppm quinine solution at 50 mM NaCl and 0.05 M H_2SO_4 .
3. Record the emission of both solutions.

Data analysis and presentation

Part 1: Objective to construct a calibration curve and determine the limit of detection

(i) Data treatment

Construct a table to show the corrected intensity values (must subtract the background emission recorded for the blank).

Calculate mean, Standard deviation and the standard error for each set of three emission values collected for each concentration.

1. Using Excel form a table that lists the emission intensity for each concentration. (Table 1)

Conc	0.02 ppm					2 ppm
^a Intensity (Sample 1)						
^a Intensity (Sample 2)						
^a Intensity (Sample 3)						
Mean ($\bar{x}$)						
SD (s)						
Standard Error (SE)						

^aIndicate that the background intensity at the maximum wavelength 450 nm has been subtracted.

To calculate mean or Average (=AVERAGE(cell number 1, cell number 2, cell number 3))

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

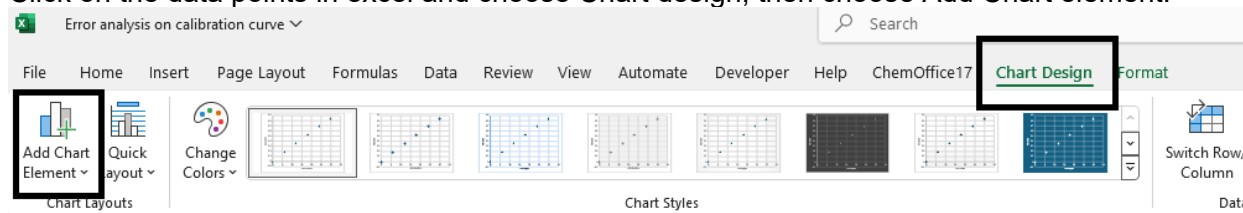
Standard Deviation (=STDEV=(first cell in a column : last cell in that column))

$$s = \sqrt{\frac{\sum_i (x_i - \bar{x})^2}{n - 1}}$$

Determine the standard error (SE) = $s/\sqrt{n}$ where n is the number of samples at each concentration (here n = 3).

Graph 1: plot the mean data point and include vertical errors bars representing the standard deviation in intensity.

Click on the data points in excel and choose Chart design, then choose Add Chart element.

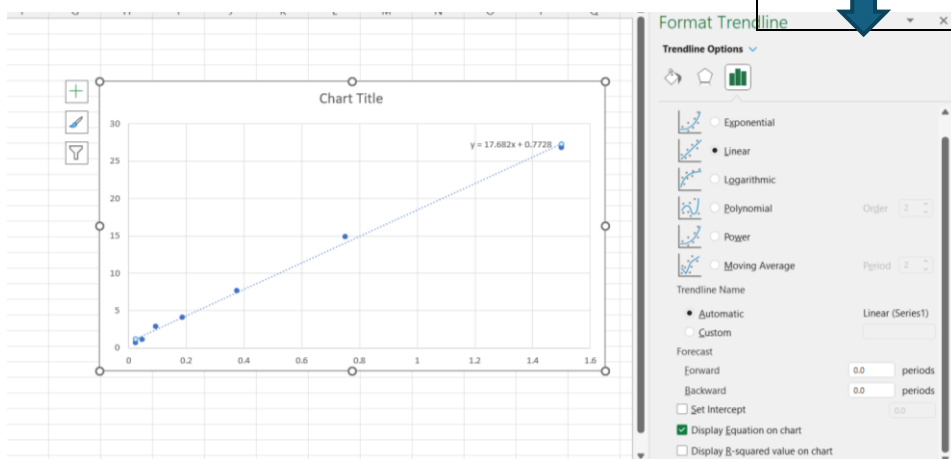


Choose insert error bars.

(ii) Data treatment and Graph 2

Generate a second table of Intensity and Concentration. This should combine the measurements from all samples. Then generate a plot containing all the data. You will have a spread of points around each concentration, and you are going to fit the best line to these points. Click on a data point and choose add trendline and then choose linear and show equation.

Conc	Intensity
2 ppm	1
2 ppm	2
2 ppm	3
1.8 ppm	1
1.8 ppm	2



2. Calculate LOD and LOQ. Need to add the trend line to the data. Apply the LINEST function in Excel extract the values for the linear regression analysis of the best fit of the line. Determine the equation of the line and obtain m .

Part 2: Determination of the concentration of quinine and the uncertainty in the value

See pages 86-88 Harris 8th Edition Quantitative Chemical Analysis.

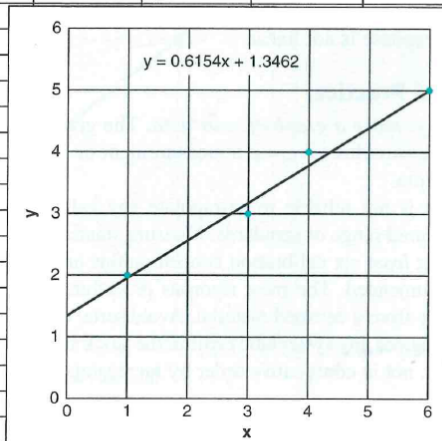
1. Using the equation of the lines from the calibration curve determine the concentration of quinine in the two tonic water samples.
2. LINEST treatment in Excel provides details of the uncertainty in the intercept, slope and the values of y . This information can be used to determine the uncertainty in the concentration s_x that is extracted. This is done using the equation below:

$$s_x = \frac{s_y}{|m|} \sqrt{\frac{1}{k} + \frac{1}{n} + \frac{(y - \bar{y})^2}{m^2 \sum_{i=1}^n (x_i - \bar{x})^2}}$$

s_y = standard deviation in y derived from LINEST	y = measured value
$ m $ = absolute value of the slope	$\bar{y}$ = mean value of y for the points on the calibration curve
k = number of replicate measurements of unknown sample	x_i = individual values of the x data points
n = number of data points in the calibration plot	$\bar{x}$ = mean value of x for the points on the calibration curve

s_x can be solved by creating the excel sheet below.

	A	B	C	D	E	F	G	H	I
1	Least-Squares Spreadsheet								
2									
3	Highlight cells B10:C12	x	y						
4	Type "= LINEST(C4:C7,	1	2						
5	B4:B7,TRUE,TRUE)	3	3						
6	For PC, press	4	4						
7	CTRL+SHIFT+ENTER	6	5						
8	For Mac, press								
9	COMMAND+RETURN	LINEST output:							
10	m	0.6154	1.3462	b					
11	s _m	0.0544	0.2141	s _b					
12	R ²	0.9846	0.1961	s _y					
13									
14	n =	4	B14 = COUNT(B4:B7)						
15	Mean y =	3.5	B15 = AVERAGE(C4:C7)						
16	Σ(x _i - mean x) ² =	13	B16 = DEVSQ(B4:B7)						
17									
18	Measured y =	2.72	Input						
19	k = Number of replicate measurements of y =	1	Input						
20	Derived x =	2.2325	B20 = (B18-C10)/B10						
21	s _x =	0.3735	B21 = (C12/B10)*SQRT((1/B19)+(1/B14)+((B18-B15)^2)/(B10^2*B16))						



Expressing the final answer and noting significant figures

Linear regression analysis of the calibration curve will provide a value of the uncertainty s_x .

Important: the first digit of the uncertainty is the last significant figure. Example, where

$$m = 0.61528 \text{ and } s_m = \pm 0.05439 \rightarrow 0.62 \pm 0.05$$

Alternatively, can retain first insignificant figure as a subscript in the answer $\rightarrow 0.61_5 \pm 0.05_4$

It is important to consider the significant figures in your answer for concentration, which should be presented in the general form: $x \pm s_x$. Include both forms shown above in your report.

In the case above inserting the value of 2.72 gives rise to an x value of 2.2325. But $s_x = 0.3735$. So the answer quoted is $2.2_3 \pm 0.3_7$. Using this approach the value for concentration of quinine can be determined from the intensity input.

Questions

1. Comment on the LOD obtained and the intensity of the blank solution.
2. Using the excel spreadsheet comment on the relative impact of n and k on the uncertainty in the concentration.
3. The emission of quinine is observed to change with pH and at increasing pH the emission is observed to decrease. Comment on the likely cause of this change.
4. What possible challenges may be encountered if the experiment was performed using tonic water sourced from a freshly opened bottle.
5. Comment on the concentration of the quinine in terms of the EU limits.
6. Discuss the possible sources of error in this experiment that may not have not been accounted for.

Acknowledgements

SJQ acknowledges the input of Mr Hans Eckhardt in the development of this lab.

References

An account of a new species of the bark-tree, found in the Island of St. Lucia', Phil. Trans. R. Soc. Lond., 1784, 74: 452–6.

Report

Your report should include:

1. A brief introduction and discussion of the background and objectives.
2. The tabulated data obtained in the experiment showing the concentrations and intensities.
3. Be sure the intensities have been corrected for the background fluorescence.
4. Part 1 Comment on the standard error determined for the different concentrations (Table 1).
5. Part 1 should include a plot of the calibration curve a representation of error bars included.
6. Part 1 should include a plot of the calibration curve with all the points included (Table 2).
7. The equation of the line extracted from the calibration curve.
8. Calculated values for LOD and LOQ.
9. Part 2: The concentration of quinine determined in each commercial sample.
10. Part 2: the uncertainty in the value of quinine determined.
11. Your calculations should be clearly shown to show the dilutions used for all standards.
12. Part 3 The plots of the emission spectrum in the absence, and presence of 50 mM NaCl solution.
13. Answers to the questions.

Note all plots should have a figure caption that details the nature of sample and the solution conditions. Each axis must be labelled in the plots.

Additional note on Errors

The statistical error will be determined by linear regression in Excel but it is also important to note at that errors may be accounted for by considering the propagation of the systematic errors associated with glassware.

Useful to comment on the uncertainty associated with glassware.

Consider 25-mL pipette that is certified to deliver 25.00 ± 0.03 mL.

Multiple deliveries from a pipette.

The range here is known as the tolerance ($a = 0.03$).

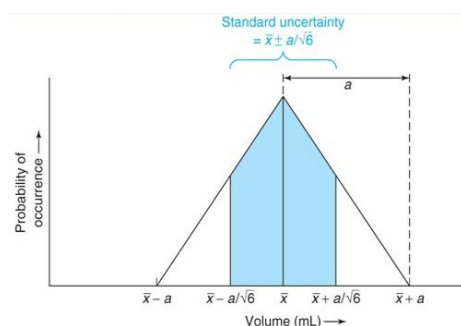
Efforts made by manufacturer to dispense close to 25.00 mL.

Highest probability the pipette will deliver 25.00 mL and then falls off in a linear manner.

Here the standard error across many pipettes is approximated by a triangular distribution.

For a **triangular distribution**, the **standard deviation** or **standard error** is:

$$\bar{x} \pm a/\sqrt{6}$$



RISK ASSESSMENT – Exp 1				Record no. AM30110
SUBSTANCE NAME	Physical Form	Quantity	NATURE OF HAZARD (from Sigma-Aldrich and TCI MSDS information)	Hazard symbol
0.05 M H ₂ SO ₄ (dilute sulfuric acid solution)	liquid	Apprx 225 mL	H290 - May be corrosive to metals	
2 ppm quinine sulphate solution	Liquid	10 mL	H290 - May be corrosive to metals	
NaCl	Solid	< 1g	Not hazardous	
NATURE OF PROCESS • Provided with a 2 ppm quinine sulphate stock solution to be diluted in 25 mL volumetric flasks with 0.05 M sulfuric acid solution to prepare a series of standards. • Determine the intensity of the quinine emission using the fluorimeter.				
Is there a less hazardous substance? No If so, why not use it? N/A				
CONTROL MEASURES REQUIRED • Good laboratory practice is to be followed at all times (i.e. lab coat and safety spectacles). • Care to be taken when using glass pipettes.				
DISPOSAL PROCEDURE Solutions are to be disposed down the sink.				
SPILLAGE Wipe up small spillages at any stage with paper towel.				
UNCONTROLLED RELEASE N/A – only small quantities used				
FIRE If small, attempt to extinguish ethanol fires with a CO ₂ extinguisher. If the fire cannot be contained, evacuate area and obtain assistance.				
FAILURE OF LOCAL EXHAUST CONTROL N/A				



PRACTICAL 2

Use of Atomic Absorption Spectroscopy to Determine the Trace Metal ion concentration of Calcium and Magnesium

Learning Objectives

Understand the sensitivity of atomic absorption for analysis by determining the limit of detection of metal ions in water samples.

Understand the requirements for generating a calibration curve.

Be able to use spreadsheets to manipulate data and plot spectra and graphs, including working with and expressing uncertainties.

Aim: to use atomic absorption to:

- (1) Generate an atomic absorption calibration curve using standards of known concentration.
- (2) Determine the Limit of Detection (LOD) and limit of Quantification (LOQ) of Mg^{2+} and Ca^{2+} .
- (3) Determine the concentration of metal ions in different water samples.
- (4) Determine the uncertainty in the concentration of metal ions.

Introduction

Atomic Absorption Spectroscopy (AA) is a vapour phase method for the determination of the concentration of atomic species. It offers the advantages of speed, sensitivity, precision and specificity, and is particularly suited for the determination of low metal concentrations. In this experiment, you will use AA to determine the calcium and magnesium content of a number of water samples.

The origins of AA lie in the observation of the Fraunhofer lines in the solar spectrum. It was subsequently shown that the D lines absent from the solar spectrum correspond to the emission wavelengths of Sodium. This led to the formulation of Kirchhoff's law; *All matter absorbs light at the wavelength at which it emits*. This law is particularly applicable to gaseous material.

Energy levels in atoms are quantized, *i.e.* energy can only be emitted and absorbed in discrete amounts, which correspond to the separation of the energy levels.

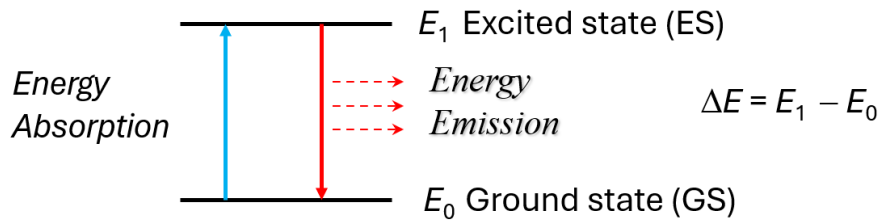


Figure 1: Absorption and emission of energy

The frequency of the radiation is directly proportional to the energy of the radiation, the constant of proportionality is Planck's constant, $h = 6.63 \times 10^{-34}$ Js

$$\Delta E = h\nu$$

As a result, atoms have characteristic emission and absorption spectra.

Atomic Absorption

In an Atomic Absorption Spectrophotometer, the solution of interest is injected as a fine spray into a flame. Typically, acetylene - air (2,150 °C) and Acetylene - Nitrous oxide (2,955 °C) are used. The light source is a hollow cathode lamp, containing the element of interest. In the interests of economy, it is possible to purchase lamps containing more than one element. For example, the lamp used in this experiment generates emission lines from both calcium and magnesium.

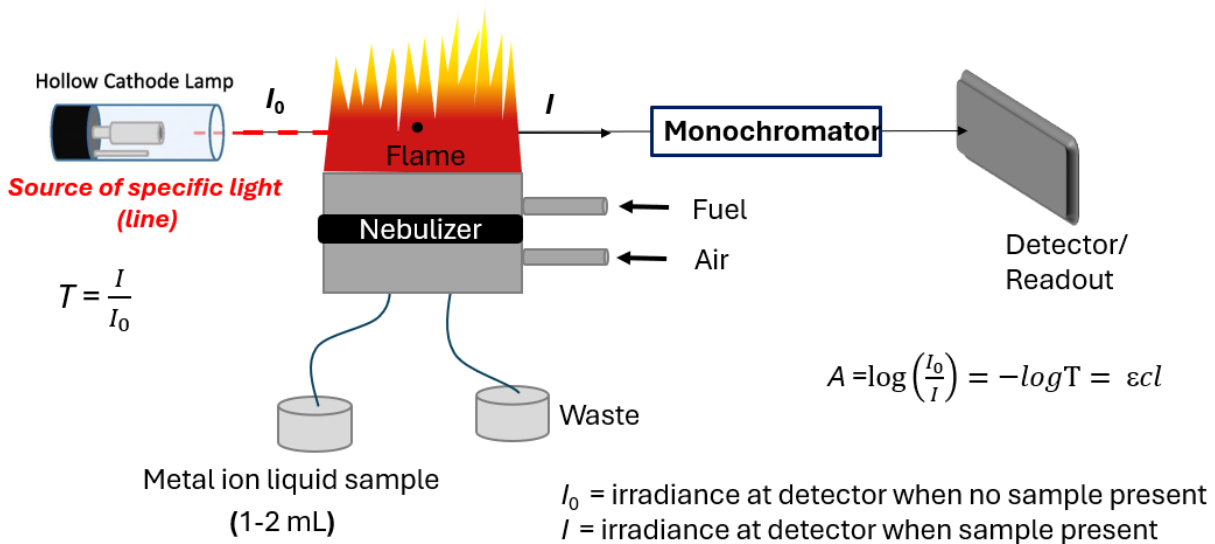


Figure 2: Schematic Diagram of an Atomic Absorption Spectrophotometer

The relationship between the amount of light absorbed and concentration is given by the Beer-Lambert law: $A = \epsilon c l$

where:

ϵ = a constant characteristic of the system (which can be thought of as the 'allowedness' of the transition),

c = concentration,

l = pathlength,

and the ratio $\log_{10}(I_0 / I)$ is termed the *absorbance* (A).

Calcium and Magnesium

Both calcium and magnesium play a vital role in human metabolism. As well as its structural role in bones and teeth, Calcium is also involved in blood clotting and muscle contraction. The European food Safety Authority recommended intake of 1000 mg/day for adults of which about 75% comes from dairy products and for children aged 1–3, 4–10 and 11–17 years of 450, 800 and 1 150 mg/day, respectively. Magnesium also plays a structural role in bones, as well as being involved in energy release and temperature regulation. For adults the magnesium adequate intake (AI) is 350 mg/day for men and 300 mg/day for women. For children aged 1 to < 3 years, an AI for magnesium is set at 170 mg/day for both sexes 0.4 g/day, of which 24% comes from grain and cereals.

Safety

The drinking water samples are intended for human consumption but may be contaminated by the laboratory environment. Acetylene forms explosive mixtures with air and is narcotic and asphyxiant in high concentrations. Follow all instructions carefully and do not adjust the instrument and gas cylinder controls. The burner can reach temperatures of up to 2,150°C. Do not touch!

Method

Calibration curve:

You are supplied with two stock solutions prepared from the chloride salts of magnesium and calcium. The first 1,000 ppm in Ca^{2+} and the second 100 ppm in Mg^{2+} (1 ppm is one part per million, *i.e.* 1 mg/L). Prepare an intermediate solution of 100 ppm by diluting 10 mL of stock solution to 100 mL and mixing thoroughly. Prepare solutions of concentrations 3, 6, 12, 18 and 24 ppm in calcium by dispensing 3, 6, 12, 18 and 24 mL of the intermediate standard respectively and making up to 100 mL. This process avoids the errors inherent in dispensing very small volumes and is known as *serial dilution*.

Likewise prepare a 10 ppm magnesium standard by diluting 10 mL of the 100 ppm standard to 100 mL. Prepare solutions of concentrations 0.2, 0.4, 0.6, 0.8 and 1.0 ppm in magnesium by dispensing 2, 4, 6, 8 and 10 mL of the intermediate standard respectively and making up to 100 mL. **Three instrument replicate measurements** will be recorded for each concentration. The outputs for these values and the mean should be tabulated.

Analysis of water samples:

Next samples of water will be analysed to determine and the concentration determined using the calibration curve.

It is possible that some unknowns will give a value greater than the top of the range for which the instrument has been calibrated. Extrapolation of calibration curves is to be avoided. The correct method is to dilute the sample by a known factor, thus bringing the measured value within the calibration range.

Tap water the calcium concentration is typically low and can be analysed without dilution. However, the magnesium content requires dilution, typically to 1 in 4 volume by taking 25 mL and diluting with de-ionised water to prepare 100 mL.

Bottled water the ion concentration is typically higher than that in tap water and so it is recommended to consider for **calcium** a 1 in 4 volume by taking 25 mL and diluting with de-ionised water to prepare 100 mL and for **magnesium** 1/40 solution by taking 2.5 mL and diluting with de-ionised water to prepare 100 mL

Your demonstrator will have prepared the instrument for the analysis of either calcium or magnesium. It should be zeroed using de-ionised water. Introduce each standard in turn into the atomiser, and record the absorbance value.

Do not allow air to be drawn into the atomised immediately after a sample measurement. This will cause solid material to deposit in the atomiser, clogging it.

Data treatment

Generate **two tables** for absorbance data obtained for the range of $[\text{Mg}^{2+}]$ and $[\text{Ca}^{2+}]$. The table should have the format:

Table x: title

Concentration ppm	Abs 1	Abs 2	Abs 3	Mean	Standard dev
0.0					
6.0					
12.0					
18.0					
24.0					
30.0					

Graphs 1 & 2: First generate a plot using the mean abs data determined from the three replicates. Comment on why it may be challenging to include the error bars represented by the standard deviation.

Graphs 3 & 4: Using all the data points obtained plot a calibration graph of absorbance vs. concentration for each element. The calibration curve should include all the measured values, unless there is a clear outlier.

Using graphs 3 & 4 perform the following analysis. Refer to the description in experiment 1 to perform the analysis.

Add a trend line to the curve and perform the LINEST analysis in excel to determine

- 1) the LOD and LOQ for $[\text{Mg}^{2+}]$ and $[\text{Ca}^{2+}]$ ions
- 2) values of $[\text{Mg}^{2+}]$ and $[\text{Ca}^{2+}]$ in each of the unknowns supplied to you and
- 3) the uncertainty in the concentration.

Errors

The statistical error will be determined by linear regression in Excel but it is also important to note at that errors may be accounted for by considering the propagation of the systematic errors associated with glassware.

Useful to comment on the uncertainty associated with glassware.



Consider 25-mL pipette that is certified to deliver 25.00 ± 0.03 mL.

Multiple deliveries from a pipette.

The range here is known as the tolerance ($a = 0.03$).

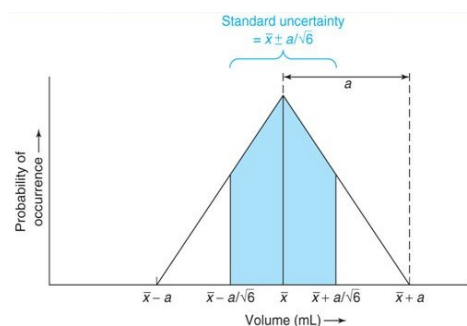
Efforts made by manufacturer to dispense close to 25.00 mL.

Highest probability the pipette will deliver 25.00 mL and then falls off in a linear manner.

Here the standard error across a large number of pipettes is approximated by a triangular distribution.

For a **triangular distribution**, the **standard deviation** or **standard error** is:

$$\bar{x} \pm a/\sqrt{6}$$



Question

Assuming that 5% of the Calcium intake and 10% of the Magnesium intake comes from drinking water, what volume of the various waters tested would you need to drink each day?

RISK ASSESSMENT – Exp 2				Record no. AM30110
SUBSTANCE NAME	Physical Form	Quantity	NATURE OF HAZARD (from Sigma-Aldrich and TCI MSDS information)	Hazard symbol
1000 ppm CaCl ₂ solution	liquid	ca. 100 mL	H319 Causes serious eye irritation	
100 ppm MgCl ₂ solution	Liquid	ca. 10 mL	Harmless	
NATURE OF PROCESS • Provided with a 1000 ppm CaCl₂ stock solution and 100 ppm MgCl₂ to be diluted in 10 mL volumetric flasks with de-ionised water to prepare a series of standards. • Determine the intensity of the ion absorbance using AAS.				
Is there a less hazardous substance? No If so, why not use it? N/A				
CONTROL MEASURES REQUIRED • Good laboratory practice is to be followed at all times (i.e. lab coat and safety spectacles). • Care to be taken when using glass pipettes.				
DISPOSAL PROCEDURE Solutions are to be disposed down the sink.				
SPILLAGE Wipe up small spillages at any stage with paper towel.				
UNCONTROLLED RELEASE N/A – only small quantities used				
FIRE If small, attempt to extinguish ethanol fires with a CO ₂ extinguisher. If the fire cannot be contained, evacuate area and obtain assistance.				
FAILURE OF LOCAL EXHAUST CONTROL N/A				

PRACTICAL 3

Analysis of Alcoholic Drinks by Gas Chromatography

- Objectives**
- Identification of the various components of a gas chromatograph (GC)
 - To learn how to use a GC
 - To observe the effect of temperature on a GC run
 - To identify components of alcoholic drinks by GC
 - To understand the use of an internal standard in analysis
 - To determine the percentage of methanol in a sample of poitin
 - To gain insight into the analytical chemistry aspects of law enforcement

Introduction

Gas Chromatography (GC) is a method of separating and analysing mixtures of gases, liquids or volatile solids. In the machine, the material to be analysed (*analyte*) is vaporised and passes through a very long (many metres), thin (micron size) heated column. It is carried through the column by an inert gas (usually helium). As the mixture passes through the column, the various components travel at different speeds and so are separated. They therefore emerge at different times (*retention* times) and can be detected separately. The most common detection (and used here) is flame ionisation where the column effluent is burned in a hydrogen/air mixture and the produced ions detected as a tiny current.

In this practical, you are given samples of a selection of low molecular weight alcohols: methanol, ethanol, n-propanol and *iso*-butanol. The GC has a DB-5 column. This is a slightly polar column (see diagram) and is a bonded phase. Although it is fairly non-polar, it is suitable for such alcohols, and is set to a particular flow rate, while its oven temperature is variable.

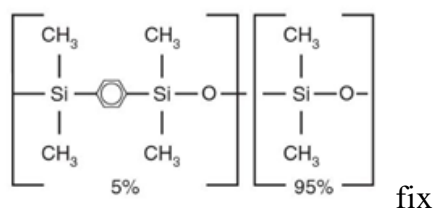


Diagram: Structure of DB-5 column: (5%-Phenyl)-methylpolysiloxane (5% of the Si atoms bear phenyl groups).

You have to determine the retention times of the alcohols on the column. You also have samples of some spiritous drinks and you have to try to identify as many components as you can in **one** of these.

You also are given a sample of poitín. Your task is to analyse it to see if it contains methanol (which is poisonous) and, if so, how much. The amount of methanol is determined by the **internal standard method** (see below), using decane as the internal standard. You are told the response factor (burn ratio) of decane to methanol. By preparing a sample of poitín containing a known concentration of decane, it is possible to relate the intensities of the methanol and decane peaks on the chromatogram and then determine the concentration of methanol in your poitín sample.

You will then give **qualitative evidence** that the peak you have assigned as methanol from poitín is actually from methanol (not another chemical with the same retention time). To do this, you will perform a *co-injection* of pure methanol and poitín.

It is recommended that you familiarise yourself with Chapter 23 from the textbook Quantitative Chemical Analysis by Daniel Harris, before attending this practical.

Safety

All of the alcohols (incl ethanol) are poisonous by ingestion, especially methanol, which causes blindness and death. However, the amounts used are very small, which mitigates the risk. More significantly, there is a substantial stick injury risk from the thin syringe needles used – be especially careful while doing the injections onto the machine. Finally two parts of the GC machine become very hot during operation (the injector block and the flame ionisation block) and therefore carry a substantial burn risk. Do not touch the autosampler while it's in operation as it can cause puncture wounds.

Procedure

You are provided with a Hamilton microlitre syringe (expensive). Please treat it with care. Before each use, draw up and expel into the solvent waste some of the liquid to be transferred. Do this three times. After use rinse out your syringe with ethanol.

- Using the Hamilton syringe, make up a 10 mL volumetric each containing. 25 μ L of a series increasing of alcohols (methanol to isobutanol) in ethanol (6 volumetrics total). Then transfer into a clean GC vial using a new glass pipette, half filling the vial and then placing a slitted vial cap on it.
- You are provided with 6 x 10 mL volumetric flasks. Using them make up solutions as follows: Flask 1. Ethanol.
Flask 2: Methanol (25 μ L) in ethanol.

Flask 3: Methanol and 1-propanol (each 25 μL) in ethanol (10 mL).

Flask 4: Methanol, 1-propanol and iso-butanol (each 25 μL) in ethanol (10 mL).

Flask 5: Methanol, 1-propanol, iso-butanol and ethyl acetate (each 25 μL) in ethanol (10 mL)

Flask 6: Methanol, 1-propanol, iso-butanol and n-decane (each 25 μL) in ethanol (10 mL).

- Using fresh glass Pasteur pipettes each time, take a sample of each of the six solutions and transfer them to six fresh GC- vials
- Take a sample of the poitín and place it into a GC-vial using a fresh glass Pasteur pipette.
- On the GC instrument with the help of your demonstrator in the OpenLab software, set up a sequence to inject from your seven vials.
- Obtain *qualitative evidence* that supports the presence of any methanol in the poitín by co-injecting it with the authentic methanol solution. Do this by adding 25 μL of methanol to a new 10 mL volumetric flask and make up to the mark with the poitín. Transfer a sample of this to a GC-vial and run (either separately or in your main sequence above). Repeat this injection at a higher or lower temperature.
- Determine the percentage of methanol by the *internal standard method*. Do this by adding 25 μL of n-decane to a new 10 mL volumetric flask and make up to the mark with the poitín. Transfer a sample of this to a GC-vial and run (either separately or in your main sequence above).
- Work out the volume percentage of methanol from the raw integration data. The relative response factor between methanol and decane is 0.33 (decane stronger).
- Dispose of any unwanted samples in the solvent waste container.

Method and Instrument Details:

The instrument is an Agilent 8860 GC-FID with a DB-5 column (30 m x 0.32 mm with a film thickness of 0.25 μm). The carrier gas is helium.

The analysis of the alcohol method parameters are as follows:

Injection volume: 0.1 μL

Inlet temperature: 250 °C

Column flow: 1 mL/min (pressure value of 4.87 psi or average velocity of 18 cm/sec): Constant flow

Split ratio of 5:1

Column Temperature: isothermal 30 °C *

Detector Temperature: 300 °C with 25 mL/min He make up flow.

*An exception to this is for the 40 °C isothermal run and the samples containing n-decane, where the oven is kept at 30 °C *for 7 minutes (normal run time) and then ramped to 120 °C at a rate of 20 °C/min holding at 120 °C for 5 minutes.

Notes on the Write Up

Your report should be typed, printed and submitted back to the laboratory **one week after** your session. Include the chromatograms that you obtained in your write up. Excluding the chromatograms, the write up should be no more than five pages and, as well as the usual title, date, etc., should cover the following points:

- Give a *very brief* introduction to GC.
- Describe the actual GC machine that you used, quoting all relevant settings.
- Discuss your results, showing how you used the data on the chromatograms to identify the various components in the alcoholic drink and the identity and percentage of methanol in the poteen.
- Discuss *in detail*, potential sources of error in the analyses that you did and ways that it might be improved.

Questions (Include answers with your write-up)

1. Why was GC the most appropriate technique for this analysis?
2. What gases were used and what was their source. What was the purpose of each gas?
3. Why is Carbowax a suitable stationary phase for this analysis?
4. What happened when the temperature of the analysis was lowered?
5. Suggest reasons for the elution order that you observed for the alcohols and decane.
6. Suggest possible identities for the additional peaks in the chromatogram. How might you try to identify these in the absence of authentic samples? Other alcohols and other isomers of the alcohols.

RISK ASSESSMENT				Record no. AM30110 GC EXP
SUBSTANCE NAME	Physical Form	Quantity	NATURE OF HAZARD (from Sigma-Aldrich and TCI MSDS information)	Hazard symbol
· Methanol	liquid	100 mL	H319 Causes serious eye irritation. H361d Suspected of damaging the unborn child. H225 Highly flammable liquid and vapour; H301 Toxic if swallowed; H311 Toxic in contact with skin; H331 Toxic if inhaled; H370 Causes damage to organs.	  
· Ethanol	Liquid	100 mL	H224 Flammable liquids (cat 2). H319 Eye irritation (cat 2).	 
· Ethyl acetate	Liquid	100 mL	H225 Flammable liquid H319 Eye irritation H336 central nervous system	 
· 1-Propanol	Liquid	100 mL	H225 Highly flammable liquid and vapour. H318 Causes serious eye damage. H336 May cause drowsiness or dizziness.	  
· Isobutanol	Liquid	100 mL	Flammable liquid (Cat 3), H226 Skin irritation (Cat 2), H315 Serious eye damage (Cat 1), H318 Specific target organ toxicity - single exposure (Cat 3), Central nervous system, H336 Specific target organ toxicity - single exposure (Cat 3), Respiratory system, H335	  
· n-Decane	Liquid	100 mL	Flammable liquids (Category 3), H226 Aspiration hazard (Category 1), H304	 
NATURE OF PROCESS – see over				

NATURE OF PROCESS	
• Provided with a microliter syringe. • Before each injection draw up solvent and expel to wash out the needle. • Inject 0.1-0.2 microliter of ethanol, then add 6-7 drops of methanol to the master ethanol sample and inject 0.1-0.2 microliter again. • Repeat for 1-propanol, isobutanol, ethyl acetate and decane. • Inject poitin sample and determine if there is any methanol present by co-injecting it with the authentic methanol solution. • Determine the percentage of methanol by following the procedure given.	
Is there a less hazardous substance?	No
If so, why not use it?	N/A
CONTROL MEASURES REQUIRED	
• Good laboratory practice is to be followed at all times (i.e. lab coat and safety spectacles). • Disposable nitrile gloves are to be worn when handling reagents and solvents. • Solvents should be drawn up using glass pipettes and disposed of in sharps bin provided at the end of the practical. • Handle syringes carefully and place back into the box when finished. Broken glassware to be reported to the technical officer for disposal and a replacement given.	
DISPOSAL PROCEDURE	
Filtrates and solvent wastes are to be disposed of in the waste flammable solvent canister. Chloroform filtrates to be disposed of in the chlorinated waste solvent canister.	
EMERGENCY ARRANGEMENTS	
Turn off lab services and close all gas cylinders.	
SPILLAGE	
Wipe up small spillages at any stage with paper towel and rinse towel through a funnel into the flammable waste solvent container. In case of a large spillage of solvent, add sand to absorb the spill, wash sand with ethanol then water through a filter funnel into the flammable solvent bottle.	
UNCONTROLLED RELEASE	
The room has 6-10 volume changes per hour and	
FIRE	
If small, attempt to extinguish ethanol fires with a CO ₂ extinguisher. If the fire cannot be contained, evacuate area and obtain assistance.	
FAILURE OF LOCAL EXHAUST CONTROL	
Services would shut down the gases so instrument would turn off. Evacuation of the lab to follow until services restored.	

PRACTICAL 4

Analysis of Soft Drinks by Liquid Chromatography

- Objectives**
- Identification of the various components of a liquid chromatograph (LC)
 - To learn how to use an LC
 - To observe the effect of flow rate on an LC run
 - To observe the influence of detector wavelength on an LC analysis
 - To understand the use of an external standard in analysis
 - To determine the concentration of caffeine in an energy drink
 - To gain insight into the analytical chemistry aspects of food regulation

Introduction

High Performance Liquid Chromatography (HPLC) is a method of separating and analysing mixtures of solids. It is one of the most common methods of analysis, especially of pharmaceutical substances. The mixture to be analysed (*analyte*) is introduced into the machine as a solution and is passed through a steel column packed with a silica-based material. It is carried through the column by a specific solvent mixture (the *mobile phase*). High back pressure develops in the column, and therefore the machine has powerful pumps to get a reasonable flow rate (mL/min). In turn this means that the solvents used have to be very pure and contain no particulates. As the mixture passes through the column, the various components travel at different speeds and so are separated. They therefore emerge at different times (*retention times*) and can be detected separately. The most common detection method is used here - ultraviolet absorption spectroscopy.

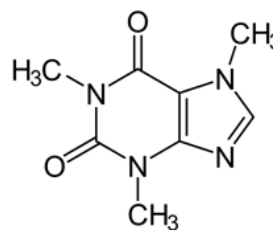
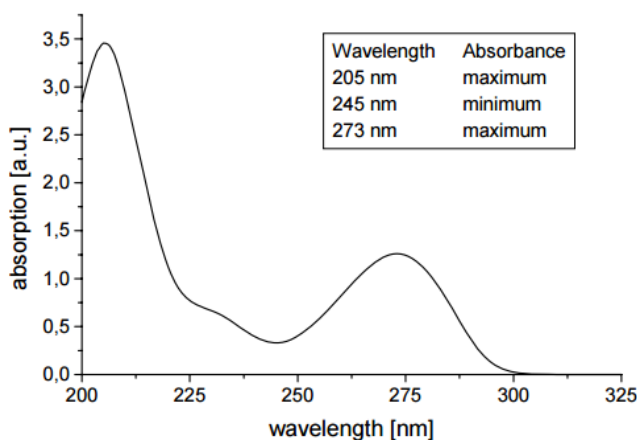
In this practical, you are going to analyse the concentration of caffeine in a sample of an energy drink. This experiment involves a quantitative part and a qualitative part. The HPLC has a Zorbax C-18 column (non-polar) and is eluted with a particular solvent mix (20% methanol / 80% 50 mM formic acid/sodium formate buffer @ pH 3.4 in water). Its flow rate is variable and its UV detector can emit/detect at different wavelengths.

First, you have to determine the retention time of caffeine on the column using an authentic sample. Then you will construct a calibration graph of concentration of caffeine (in mg/mL) versus integration of the caffeine peak on the chromatogram (in AU², arbitrary units squared). You will then use this to measure (i.e. **quantitate**) the amount of caffeine in the energy drink sample using the **external standard** method.

You will then give **qualitative evidence** that the peak you have assigned as caffeine from the energy drink is actually from caffeine (not another chemical with the same retention time). To do this, you will perform a *co-injection* of the pure caffeine sample and the energy drink.

It is recommended that you familiarise yourself with Chapter 23 from the textbook Quantitative Chemical Analysis by Daniel Harris, before attending this practical. Pay particular attention to the **van Deemter equation**, as this will make you well-equipped to understand the factors that affect the separation of analytes on a liquid chromatographic column.

UV/VIS Spectrum Caffeine in water (concentration 25 µg/ml, path length 1 cm)



Safety

The energy drink samples are intended for human consumption but may be contaminated by the laboratory environment. The solvent mixture (*mobile phase*) contains methanol, which is highly poisonous by ingestion (causes blindness and death). The mobile phase is also somewhat acidic and is an irritant. Wear suitable protective gloves and avoid breathing its vapour. There is the usual stick injury risk from using pipettes.

Procedure

You are provided with glass graduated pipettes (1 mL). After each use, rinse out with HPLC grade water and leave aside to dry. If you need to quickly dry, do a further rinse with 2-propanol.

- You are provided with a stock solution containing $1.000 \text{ g} \pm 0.001 \text{ g}$ caffeine in 100 mL of mobile phase solvent mixture, referred to as the MASTER STOCK solution (10 mg / mL).
- Using a glass graduated pipette, take 1 mL of the master stock solution and dilute it with mobile phase mixture by transferring it and making up to the mark in a 100 mL volumetric flask. This is your STUDENT STOCK solution (0.1 mg / mL).

- Using appropriate dilutions of your student stock solution, make up 4 more standard solutions containing the following concentrations: 0.05 mg/mL, 0.01 mg/mL, 0.005 mg /mL and 0.001 mg /mL.
- Make up a sample of your energy drink by transferring 1 mL of it and making up to the mark in a 10 mL volumetric flask. Invert 10 times.
- Using fresh glass Pasteur pipettes each time, take a sample of each of the six solutions and transfer them to six fresh HPLC-vials.
- Make up a solution containing 50% of your energy drink sample and 50% of your 0.05 mg/mL standard. This is your co-injection sample.
- Make sure to get a caffeine LRM (Lab Reference Material) sample from your demonstrator (located in the freezer). This is a sample that every group (this year) will run and be used to calculate internal variance between samples run on different dates/times.
- You will also need a blank sample for the first injection.
- With the help of your Demonstrator, create a sequence on the HPLC computer containing the blank, each of the five standards, the energy drink solution, the LRM and the co-injection. Use the data obtained for the standard solutions to construct a calibration curve of concentration (in mg/mL) versus peak area (in AU²).
- The sequence you create should also contain a case where, using the 0.05 mg / mL standard, a 20% higher flow rate is used compared to the normal flow rate (1 mL/min).
- The sequence you create should also contain a case where, using the 0.05 mg / mL standard, the detector wavelength is changed from the standard wavelength (254 nm) to a longer wavelength (270 nm).
- Using AU² versus concentration calibration curve (**quantitatively**), work out the concentration of the caffeine in the energy drink. This procedure is called the *external standard* method.
- Clean and dry all glassware; make sure to rinse the pipettes with HPLC water and then 2-propanol and dispose of any unwanted samples in the solvent waste container.

Standard Method and Instrument Details:

The HPLC is an Agilent 1200 with a DAD (Diode Array Detector - a type of UV/vis detector). The column is an Agilent Zorbax C18 (150 mm x 4.6 mm with 5 μ m silica particles) with an isocratic eluent of 20% methanol 80% 50mM sodium formate at pH 3.7 at a flow rate of 1 mL/min. the wavelength of detection is 254 nm as standard. The standard injection volume is 20 μ L.

Notes on the Write Up

Your report should be typed, printed and submitted back to the laboratory **one week after** your session. Include the chromatograms that you obtained in your write up. Excluding the chromatograms, the write up should be no more than five pages and, as well as the usual title, date, etc., should cover the following points:

- Give a *very brief* introduction to HPLC.
- Describe the HPLC machine that you used, quoting all relevant settings and include a diagram.
- Discuss your results, showing how you used the data on the chromatograms to determine the identity and percentage of caffeine in the energy drink. Make sure to include properly labelled graphs and tables of all your data.
- Discuss *in detail*, potential **sources** of error in the analyses that you did and ways that it might be improved.

Questions (Include answers with your write-up)

1. Why was liquid chromatography the most appropriate technique for this analysis?
2. Why is C-18 a suitable stationary phase for this analysis?
3. What solvents were used and what was the purpose of each?
4. What happened when the flow rate was raised? Discuss, in your own words, how this affected separation of the analytes (refer to the van Deemter equation).
5. How did the peaks on the chromatograms differ between the detector channels set at 254 nm and 270 nm? Was there a difference between the two groups' results?
6. There are a number of additional peaks in the chromatogram. Are these compounds more or less polar than caffeine? Suggest possible identities for these peaks. Do you think these compounds would be easy to analyse by LC?
7. Was the percentage of caffeine that you measured in reasonable agreement with the manufacturer's quoted percentage? Describe the difference between accuracy/precision and how each could be improved for this experiment.
8. You performed a co-injection of caffeine with your energy drink to give qualitative evidence for the presence of caffeine. How could this evidence be unreliable? Describe some other analytical techniques shown to you during the practical and how they could give more supporting evidence for the presence of caffeine in the energy drink.

RISK ASSESSMENT				Record no. AM30110 HPLC EXP
SUBSTANCE NAME	Physical Form	Quantity	NATURE OF HAZARD (from Sigma-Aldrich and TCI MSDS information)	Hazard symbol
Methanol	Liquid	2.5 L (attached to the HPLC)	H319 Causes serious eye irritation. H361d Suspected of damaging the unborn child. H225 Highly flammable liquid and vapour; H301 Toxic if swallowed; H311 Toxic in contact with skin; H331 Toxic if inhaled; H370 Causes damage to organs.	  
Sodium formate buffer (50 mM at pH 3.7)	Liquid	1 L (attached to the HPLC)	Not a hazardous material according to Regulation (EC) No. 1272/2008.	N/A
CHEM30110 eluent (20% methanol / 80% sodium formate buffer)	Liquid	500 mL (using 100mL)	H319 Causes serious eye irritation. H361d Suspected of damaging the unborn child H301 Toxic if swallowed; H311 Toxic in contact with skin; H331 Toxic if inhaled; H370 Causes damage to organs.	 
Caffeine (10g / L)	Liquid	100 mL (using 1 mL)	H302 Acute oral toxicity (cat 4)	
NATURE OF PROCESS - Start mobile phase flow at 1 ml/min with detector at 270 nm. - Stock solution prepared 1 ml of caffeine stock in 100 mls - Inject the 10 mg/ml stock (25 µL). - Prepare a series of dilutions using a burette with mobile phase. - Create a calibration curve of conc. caffeine vs. peak area. - Take 1 ml of Red Bull, dilute to 10 mls with mobile phase and inject (30 min run). - Calculate the concentration of Caffeine in red bull. - Put 5 mls of Red bull and 5 mls of 10 mg/ml Caffeine into 10 ml volumetric and inject.				
Is there a less hazardous substance? No If so, why not use it? N/A				

CONTROL MEASURES REQUIRED – see over**CONTROL MEASURES REQUIRED**

- Good laboratory practice is to be followed at all times (i.e. lab coat and safety spectacles).
- Disposable nitrile gloves are to be worn when handling reagents and solvents.
- Solvents and solutions should be drawn up using graduated glass pipettes with a pipette filler (placed on in the correct manner as demonstrated to you).
- Handle syringes carefully and place back into the box when finished.
- Broken glassware to be reported to the technical officer for disposal and a replacement given.

DISPOSAL PROCEDURE

Filtrates and solvent wastes are to be disposed of in the waste flammable solvent canister. Any chlorinated waste to be disposed of in the chlorinated waste solvent canister.

EMERGENCY ARRANGEMENTS

Turn off HPLC at the sockets and turn off all lab services.

SPILLAGE

Wipe up small spillages at any stage with paper towel and rinse towel through a funnel into the flammable waste solvent container. In case of a large spillage of solvent, add sand to absorb the spill, wash sand with ethanol then water through a filter funnel into the flammable solvent bottle. Acid spills to be neutralized prior to clean up.

UNCONTROLLED RELEASE

6-10 air volume changes in the room per hour, plus activated charcoal filters on the waste containers.

FIRE

If small, attempt to extinguish ethanol fires with a CO₂ extinguisher. If the fire cannot be contained, evacuate area and obtain assistance.

FAILURE OF LOCAL EXHAUST CONTROL

Evacuate the lab until ventilation restored.